

Smart Meter Overview

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Knowledge Base: Smart Meter AI Assistant — Synthetic Business Document

Purpose and scope

Smart meters measure electrical energy and communicate operational data to utility systems. The assistant knowledge base uses this document for basic smart-meter concepts.

Components

A typical meter includes a metering element, communication module, display, memory, clock, and event capability. The exact hardware depends on the approved meter model.

Meter data

Important data can include cumulative energy, interval readings, voltage, current, power factor, maximum demand, billing registers, and event information.

Communication

Meter data is commonly transmitted through a communication network to HES. HES handles communication and commands, while MDM validates and processes meter data.

Remote operations

Remote operations may include remote connect, remote disconnect, meter reads, load-profile requests, time synchronization, and approved configuration operations.

Health monitoring

Operational monitoring should consider communication freshness, missing intervals, command response, electrical indicators, and meter events.

Operational note

This is synthetic content created for local RAG development, testing, interview demonstration, and GitHub publication. It is not a confidential utility procedure or contractual document.